

The Fine-Structure Constant as the Norm of a Two-Channel Projection Tensor

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Abstract

The number $\alpha^{-1} \simeq 137.036$ is, in the Standard Model, a measured input: nothing in the framework forces its value. I argue that this is not an accident of our ignorance but a structural feature of any theory written on $\mathbb{R}^4$, and that the missing structure is supplied by a single non-invertible projection from a five-dimensional bulk onto the observable boundary. The central claim of the paper is sharper than a numerical coincidence. The projection does not deliver a scalar coupling; it delivers a *two-component object* $\alpha = (\alpha_{\parallel}, \alpha_{\perp})$, whose two entries are the coherent (gravitational) and radiative (electromagnetic) channels of one conserved Noether current. What experiments call “the fine-structure constant” is the hyperbolic invariant $\alpha^{-1} = \xi\eta$ of this tensor — the one scalar that the breathing dynamics conserves exactly. A measurement is a *contraction* of α with an apparatus-dependent channel weight, so the experimental determinations of α are predicted to form a spectrum rather than a point. The self-readout class (electron $g-2$, CODATA) sits on the fixed point; the two photon-recoil determinations, which contract the anisotropic channel, are displaced from it by exactly the single-channel breathing rate, and their mutual splitting matches that rate at 0.06σ with no adjustable parameter. The pole of the boundary filter is not a fitted number: the Pincherle dilatation $\mathcal{D} = x d/dx$ of the quadratic autosimilar level, times the spectral curvature $\kappa = \sqrt{5}$, fixes the bulk-to-boundary scale $\Lambda = 2\varphi^2\sqrt{5}$, and the static pole is $\alpha^{-1} = \Lambda^2 = 20\varphi^4$ (the Fibonacci form $5 + 3\sqrt{5}$ emerging as a consequence). Two structural corrections, both parameter-free, the cycle-average of that same dynamics, bring the conserved norm to $\alpha^{-1} = 137.035\,999\,1703$, a deviation of -6.73×10^{-9} (0.32σ) from CODATA 2022, inside the experimental uncertainty. The same root matrix M whose squared norm gives α^{-1} has a diagonal contraction that gives $\sin^2 \theta_W = 3/13$ at tree level, closing to $0.231\,25$ under the same two corrections. A standalone script verifies every exact identity and every counterfactual.

Keywords: fine-structure constant; non-invertible projection; Majorana spinor; coupling tensor; hyperbolic invariant; Hilbert quadrature; fundamental constants.

1 The problem, and the shape of its solution

Sommerfeld introduced α in 1916 as the ratio of the electron’s velocity in the first Bohr orbit to the speed of light [1]. A century later its value is still something we measure and insert by hand. The Standard Model uses α in every electromagnetic amplitude but contains no equation that fixes it. This is usually filed under “things a deeper theory should explain.” I want to argue that the reason is structural and can be stated precisely: a field theory formulated intrinsically on $\mathbb{R}^4$ cannot, even in principle, contain a criterion that selects which of its own degrees of freedom are physically observable, and without such a criterion no dimensionless ratio is determined from within. The constant enters as an input because the structure that would make it an output lives outside $\mathbb{R}^4$.

If that diagnosis is right, it also tells us what to build. We need a map from a larger domain onto the observable boundary that is genuinely lossy — non-invertible, with a kernel carrying degrees of freedom eliminated *before* observation, not merely gauge-fixed away. The question this paper answers is: what is the minimal such structure consistent with one conserved flow, no pre-existing metric, and a non-trivial boundary algebra, and what does it say about α ?

The answer has a feature I want to put at the centre rather than discover at the end. Because the projection is not invertible, the conserved current splits on the boundary into two channels that cannot be recombined into a single observable. The natural object carrying the electromagnetic coupling is therefore not a scalar but a two-component tensor

$$\alpha = (\alpha_{\parallel}, \alpha_{\perp}), \quad \alpha_{\parallel} \propto \xi \text{ (curvature, mass, gravity)}, \quad \alpha_{\perp} \propto \eta \text{ (phase, radiation, EM)}. \quad (1)$$

The two channels live on the hyperbolic surface $\Sigma : \xi\eta = \alpha^{-1}$, and the scalar $\alpha^{-1} = \xi\eta$ is the invariant of the hyperbolic flow that exchanges them. That is the object we have been calling the fine-structure constant. A laboratory determination does not see α ; it sees a scalar contraction of it, weighted by how the apparatus couples to the two channels. The central physical prediction of the construction follows immediately: determinations of α that contract the invariant symmetrically (the electron $g-2$) must land on the fixed point, while determinations that contract the channels asymmetrically (photon recoil) must sit off it, by an amount set by the breathing rate of the boundary. We will see that the two most precise recoil experiments, which currently disagree with each other at 5.5σ , straddle the fixed point and split by exactly the predicted single-channel rate.

The plan is as follows. Section 2 makes the no-go argument precise: no admissibility projection exists inside an $\mathbb{R}^4$ theory, with the sharp version stated in the C^* -algebraic category. Section 3 builds the minimal projection forced by the three requirements and shows that every ingredient — the bulk manifold $M^5 = \mathbb{R}^4 \times S^1(\tau)$, the Majorana carrier $\Xi = C\bar{\Xi}$, the golden scale φ , the rank $r = 5$ — is a consequence, not a choice. The same section derives the two-channel boundary balance and the breathing dynamics, and exhibits the coupling tensor α as the natural observable. Section 4 fixes the rank and the static pole; Section 5 computes the two structural corrections and, crucially, the readout spectrum of α that confronts the recoil data. Section 6 shows that the same root matrix organises the electroweak mixing angle. Section 7 states one prediction and the open extensions.

Throughout, I keep a strict ledger. A continuous bulk — one conserved phase on $S^1(\tau)$, one self-similar scale φ , one Majorana spinor — generates under restricted projection something irreducibly discrete: five modes, the catalogue $N_{\text{adm}} = \{2, 3, 5, 7, 11\}$, an integer matrix M , a value $\alpha^{-1} \in \mathbb{Z}[\sqrt{5}]$. The discrete structure is not imposed from outside; it is the residue of a coherence that cannot cross the projection interface intact. What the bulk could not carry through is what we measure.

2 Why α cannot be derived inside $\mathbb{R}^4$

The claim of this section is a no-go theorem, and it is worth stating cleanly because the whole construction is motivated by it. Modern field theories tell us how quantities evolve once we have decided which degrees of freedom are physical. They do not contain an *intrinsic* rule for that decision; observables are taken to be whatever is formally definable and assumed reachable with enough experimental care. That assumption is never examined structurally.

2.1 What an intrinsic admissibility criterion would have to be

Suppose such a rule existed. It would be a map $E : \mathcal{A} \rightarrow \mathcal{A}_{\text{adm}}$ from the full observable algebra to an admissible subalgebra, satisfying four conditions at once: $\mathcal{A}_{\text{adm}}$ is a subalgebra; E is idempotent, $E \circ E = E$; E has a non-trivial kernel, $\ker E \neq \{0\}$; and that kernel consists of

degrees of freedom physically removed prior to observation, not merely gauge-redundant. The last condition is what separates a genuine admissibility criterion from a gauge fixing: a gauge fixing discards descriptive redundancy and keeps all physical content.

A map with a non-trivial kernel is non-invertible — and this is forced, not optional. If E were invertible, every element of $\mathcal{A}$ would be recoverable from $\mathcal{A}_{\text{adm}}$ through E^{-1} , so in the idealised limit of complete measurement the supposedly inadmissible degrees of freedom would become accessible, contradicting their inadmissibility. Restricted observational access means exactly this: the boundary observer cannot reconstruct bulk invariants from boundary data, because the bulk-to-boundary map is not invertible.

2.2 No mechanism inside $\mathbb{R}^4$ supplies it

Run through the candidates an $\mathbb{R}^4$ theory actually offers. Gauge fixing selects one representative per orbit; its kernel is descriptive redundancy, so the fourth condition fails. Coarse-graining trades fine degrees of freedom for effective ones, but the elimination is practical, not principled — in the limit of full control the degrees of freedom return. Decoherence suppresses coherences through environmental coupling, but it is a dynamical process tied to a specific environment, not a fixed idempotent on $\mathcal{A}$, so the second condition fails. Renormalisation re-expresses the theory across scales without ever identifying degrees of freedom inadmissible at *all* scales. Algebraic QFT does specify observable algebras, but through extra axioms — locality, Haag duality — that are imposed, not derived from the dynamics. None of these is an intrinsic criterion.

Theorem 1 (Non-derivability). *Let $\mathcal{A}$ be the C^* -algebra of observables of a field theory formulated intrinsically on $\mathbb{R}^4$. No map $E : \mathcal{A} \rightarrow \mathcal{A}_{\text{adm}}$ satisfying the four admissibility conditions can be obtained from the internal structure of the theory.*

Proof. The internal mechanisms above each fail one condition, as just shown. The only remaining internal possibility is an automorphic selection $\phi : \mathcal{A} \rightarrow \mathcal{A}$. But every $*$ -automorphism of a C^* -algebra is isometric: the C^* -identity $\|a^*a\| = \|a\|^2$ gives $\|\phi(a)\| = \|a\|$ for all a , hence $\ker \phi = \{0\}$ [2]. The third condition is violated. No internal mechanism satisfies all four. $\square$

Corollary 1. *A dimensionless constant is a structural invariant of $\mathcal{A}$ only if the theory internally fixes the admissible subalgebra in which the invariant is evaluated. By Theorem 1 no such subalgebra is internally specifiable on $\mathbb{R}^4$. Therefore no dimensionless coupling, α included, is derivable from within an $\mathbb{R}^4$ theory.*

This is not a complaint about missing cleverness; it closes a door. If any dimensionless constant of nature is to have a structural origin, the theory must contain a non-invertible map from its full algebra to an admissible subalgebra with a non-trivial physical kernel, and no theory built intrinsically on $\mathbb{R}^4$ contains one. What must exist is a structure that supplies the map from outside: a projection Π from a larger domain, irreversible and non-invertible, whose kernel carries the physically eliminated degrees of freedom. The rest of the paper takes this seriously and asks for the minimal honest realisation, under three requirements:

- (i) a single globally conserved flow — one Noether current, not two;
- (ii) no pre-existing metric — the observable metric of $\mathbb{R}^4$ must be an output;
- (iii) a non-trivial boundary algebra — the projection must produce observables, not just a quotient.

Within that construction α^{-1} is not a parameter. It is the unique stable operating point of the boundary filter forced by the finite rank of the minimal projection.

3 The minimal construction and its two channels

I build the structure forced by (i)–(iii), then show that the boundary it produces is intrinsically two-channel. The coupling tensor α is not added later; it is what the non-invertible projection leaves on the boundary.

3.1 The zero mode and the bulk

Requirement (i) is one globally conserved flow. A conserved current J^I must recognise itself after transport around any closed path; for $\partial_I J^I = 0$ to be self-consistent under restricted access, the current has to carry a compact internal phase, a zero mode $\chi \in S^1$, that closes coherently. This is the minimal input compatible with a conserved flow under restricted accessibility; everything else follows from it together with (i)–(iii).

The zero mode needs a domain housing one compact phase and four extended directions of observable bilinears. Four extended directions are required by (iii): fewer cannot support a non-trivial boundary algebra, and a compact extended domain would force global identifications that violate (iii). Extra compact directions beyond S^1 introduce independent conserved channels and violate (i); non-trivial fibrations couple kernel to boundary and violate (ii). By exhaustion the minimal domain is

$$M^5 = \mathbb{R}^4 \times S^1(\tau). \quad (2)$$

The coordinate τ parametrises transport around the fibre; it is *not* the physical time of $\mathbb{R}^4$, which will emerge at the boundary as the irreversible ordering of the phase between the two channels. The same $\mathbb{R}^4 \times S^1$ appears in Kaluza–Klein theory, but the logic is inverted: there the bulk metric is input and gauge fields are output, whereas here (ii) forbids any pre-existing metric and the observable metric of $\mathbb{R}^4$ is a stability condition of the lossy boundary.

3.2 The carrier is a Majorana spinor

What field lives on M^5 ? It must carry the zero mode coherently around $S^1(\tau)$, carry no net bulk charge (a single conserved flow, (i), admits no independent charge sector), and have non-trivial holonomy so that Π produces observables (iii). A single-valued carrier has trivial holonomy and is excluded by (iii); a vector field needs a metric and is excluded by (ii); a Dirac spinor carries net charge, $\psi \neq C\bar{\psi}$, and is excluded by (i); Weyl spinors are globally incomplete on $S^1(\tau)$ and cannot close the single-flow self-reference. The unique survivor is the Majorana spinor,

$$\Xi = C\bar{\Xi}, \quad \hat{H} = -1 \quad (2\pi\text{-holonomy}), \quad (3)$$

real, globally neutral, double-valued: the self-conjugacy a single conserved flow under restricted access demands. The minimal construction is therefore

$$M^5 = \mathbb{R}^4 \times S^1(\tau), \quad \Xi = C\bar{\Xi}, \quad \Pi = \int_{S^1} d\tau, \quad (4)$$

and from here everything is consequence.

3.3 The self-similar scale

The fibre supports a tower of modes. For the tower to be self-consistent under projection — each level identical to every other under rescaling — the ratio x of adjacent modes must satisfy $x^2 = x + 1$, whose unique positive root is

$$\varphi = \frac{1 + \sqrt{5}}{2}. \quad (5)$$

This is not a parameter; it is the only fixed point of $x \mapsto 1 + 1/x$ on the positive reals, the ratio any self-similar tower must carry. Transporting Ξ along $S^1(\tau)$ extends the Clifford algebra $Cl(1, 3)$ to $Cl(4, 1)$; the minimal extension introduces the involution Γ_6 with $\Gamma_6^2 = 1$, $\{\Gamma_6, \gamma^\mu\} = 0$, and chiral projectors $P_\varphi^\pm = \frac{1}{2}(1 \pm \Gamma_6)$. Coherent closure of the two branches forces the holonomy amplitude to equal φ , giving the spectral curvature

$$\kappa = \sqrt{5} = \varphi + \frac{1}{\varphi}, \quad \gamma_6 := \kappa \Gamma_6, \quad (\gamma_6)^2 = 5, \quad \Gamma_\tau = \frac{\varphi}{\sqrt{5}}. \quad (6)$$

3.4 Fibonacci occupancy is a theorem

The Majorana condition $\Xi = C\bar{\Xi}$ kills all cross-terms in the norm of the temporal mode occupancies $O_n = \langle \Xi, v_n \rangle$, leaving a second-order recursion. Its coefficients are fixed completely.

Proposition 1 (Majorana forces $F_r = r$, unique $r = 5$). *Self-conjugacy reduces the occupancy norm to $|O_{n+2}|^2 = a|O_{n+1}|^2 + b|O_n|^2$. Scale-free closure forces $a = b$; holonomy integrality on $S^1(\tau)$ forces $2a^2 \in \mathbb{Z}$, hence $a \in \mathbb{Z}_{\geq 1}$; minimality (no two bulk modes identified on the boundary) forces $ab \leq 1$, hence $a = 1$; unit boundary normalisation fixes $b = 1$. Therefore $|O_n|^2 = F_n$. For the rank to equal its own holonomy norm one needs $F_r = r$; among positive integers $F_1 = 1$ (trivial) and $F_5 = 5$ are the only solutions, with $F_n > n$ for all $n > 5$. The non-trivial rank is $r = 5$, and its primality (5 is the unique prime with $F_p = p$) rigidifies the selection.*

Proof. $F_n \sim \varphi^n / \sqrt{5}$; $F_6 = 8 > 6$ seeds an induction $F_{k+1} = F_k + F_{k-1} > 2k - 1 > k + 1$ for $k \geq 3$, so $F_n > n$ for $n \geq 6$, while $F_2, F_3, F_4 < n$. The solutions of $F_r = r$ are $r = 1$ (zero-dimensional boundary) and $r = 5$. $\square$

Fibonacci structure is thus the spectral algebra of the Majorana chiral projector on $S^1(\tau)$, not an assumption. The transport operator generating it is the companion matrix $T_F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, whose characteristic polynomial $\lambda^2 - \lambda - 1 = 0$ has roots $\varphi, -1/\varphi$, the eigenvalues of γ_6 in scale space.

Proposition 2 (Uniqueness of the minimal construction). *Under (i)–(iii) the triple (M^5, Ξ, Π) of (4) is unique up to diffeomorphism of the fibre.*

Proof. Manifold: (i) forces exactly one compact direction; (iii) forces at least four extended ones; non-trivial fibrations violate (ii). Field: single-valued carriers, vector fields, Dirac and Weyl spinors are each excluded as above, leaving $\Xi = C\bar{\Xi}$. Projection: the rank is $r = 5$ by Proposition 1, the scale is φ , the curvature $\kappa = \sqrt{5}$. No freedom remains. $\square$

3.5 The kernel, and why the boundary has two channels

Imposing $\partial_I J^I = 0$ on this structure decomposes it as $\partial_\tau J^\tau + \partial_\mu J^\mu = 0$. Restricted accessibility requires $\ker \Pi \neq \{0\}$: the zero mode cannot cross the interface completely. This is the physical heart of the construction, and it is what makes the coupling a tensor rather than a scalar.

Since Π is not isometric, the longitudinal component J^τ and the transverse components J^μ map to *distinct* boundary sectors. The global conservation law becomes a local, directional balance,

$$\partial_\xi J^\xi = -\nabla_\eta J^\eta \quad (7)$$

which does not say each side vanishes. It says that whatever the longitudinal channel ξ gains as coherent inflow, the transverse channel η must release as radiative outflow, and conversely. Name the channels by what they carry: ξ is coherent inflow — curvature, mass, gravitational binding; η is phase outflow — radiation, electromagnetic diffusion. Gravity and electromagnetism are not two forces joined by an ansatz here. They are the two compensating faces of one Noether

current, and because Π is non-invertible they *cannot* be identified into a single observable. Any object built from both is intrinsically two-component.

There is a clean signal-theoretic statement of the same fact. The projection extracts the real part of the bulk field on $S^1(\tau)$, so the boundary field is the real part of an analytic signal and the kernel is its Hilbert quadrature:

$$\Psi_{\text{bdy}} = \text{Re}(\Xi), \quad \Xi_{\text{ker}} = -i \mathcal{H}[\Psi_{\text{bdy}}], \quad \Xi = \Psi_{\text{bdy}} + i \Xi_{\text{ker}}, \quad (8)$$

with $\mathcal{H}$ the Hilbert transform on the fibre. The kernel spinor is not an arbitrary hidden sector: it is the quadrature of the observable field, physically active but not directly readable. The non-linear return term that stabilises the boundary, derived below, is precisely the back-action of this hidden quadrature. In the bulk $D_{\text{bulk}}\Xi = 0$ is linear and timeless — the analogue of the Wheeler–DeWitt constraint — and physical time appears only at the boundary, as the irreversible advance of $\arg(\xi/\eta)$.

3.6 The boundary equation as closure of the balance

The boundary field $\Psi = \Pi\Xi$ inherits no closed linear equation, because $\ker \Pi \neq \{0\}$. Its equation is the Euler–Lagrange equation of the projected action. The bulk Lagrangian $\mathcal{L} = \bar{\Xi}\gamma^A D_A \Xi - (4\alpha)^{-1} F_{AB} F^{AB} + \mathcal{L}_{\text{curv}}$ carries a curvature coupling whose coefficients are fixed by three bulk coherence conditions on the zero mode,

$$(2\alpha\xi)^2 = e/2, \quad (2\alpha\eta)^2 = \varphi^2 \cdot 10^{-4}, \quad \sqrt{\xi\eta/3} \approx \pi. \quad (9)$$

These fix the local Dirac amplitudes and the boundary metric normalisation; they do *not* redetermine the final invariant $K_\tau = \xi^* \eta^* = \alpha^{-1}$, which is fixed by the rank-5 spectral pole of Section 4. (A note on e : the Euler number is not in $\mathbb{Z}[\sqrt{5}]$; it enters (9) only as the near-identity $e \approx 2\sqrt{2}$ at the level of the spinorial lever hierarchy $\{2^{k/4}\}$, and α does not depend on this approximation. The braid-theoretic reading of that hierarchy is deferred to separate work.)

Varying the action gives the unified non-linear Dirac equation

$$(\gamma^\mu D_\mu + \alpha\xi + i\alpha\eta\gamma^5)\Psi = \frac{\delta S_{\text{ker}}[\Psi]}{\delta \bar{\Psi}}, \quad S_{\text{ker}}[\Psi] = -\text{tr}_{\text{ker}} \Pi(\rho_\Psi \log \rho_\Psi), \quad (10)$$

with $\rho_\Psi = \text{tr}_{\text{Im } \Pi} |\Xi\rangle\langle\Xi|$ the reduced density matrix on the kernel sector. The chiral structure of the two terms is not cosmetic: η carries $U(1)$ phase (transverse, imaginary, radiative) and rides the γ^5 ; ξ carries $SU(2)$ curvature (longitudinal, real, massive). Their distinction is the same Clifford distinction $P_\varphi^\pm$ that defines the two channels. The right-hand side is the return tension $V'(|\Psi|) = \delta S_{\text{ker}}/\delta |\Psi|$ — the force with which the kernel pulls the boundary field back toward the bulk coherence it could not fully carry. In the linear limit (10) reduces to the generalised chiral Dirac equation; the non-linear term is the back-action of the zero mode left behind.

3.7 The hyperbolic flow and the coupling tensor

The non-linear equation (10) has a stable fixed point Ψ^* . Linearising, the tangent operator has exactly five eigenmodes $v_0, \dots, v_4$: the rank-5 code subspace of the projection (the Mellin-diagonal closure is proved in Appendix C). At the fixed point the two channels obey

$$\dot{\xi} = -\Gamma_\tau \xi, \quad \dot{\eta} = +\Gamma_\tau \eta, \quad (11)$$

the unique dynamics of the Pincherle dilatation $D = x\partial_x$ on the two self-similar channels at rate $\Gamma_\tau = \varphi/\sqrt{5}$, projected to the boundary. In logarithmic coordinates the flow is uniform; in the (ξ, η) plane it is a hyperbolic rotation. The product is conserved *exactly*, at every instant and not only at the fixed point:

$$\frac{d}{dt}(\xi\eta) = \dot{\xi}\eta + \xi\dot{\eta} = -\Gamma_\tau \xi\eta + \Gamma_\tau \xi\eta = 0, \quad K_\tau = \xi\eta = \alpha^{-1}. \quad (12)$$

This is the structural statement of the paper, so let me make it explicit. The boundary carries a two-component coupling

$$\boxed{\boldsymbol{\alpha} = \frac{\alpha^{(G)}}{\sqrt{K_\tau}} \begin{pmatrix} \xi & 0 \\ 0 & \eta \end{pmatrix}, \quad \alpha_{\parallel} \propto \xi, \quad \alpha_{\perp} \propto \eta, \quad \alpha^{-1} = \xi\eta = \text{hyperbolic invariant.}} \quad (13)$$

The breathing flow (11) is the one-parameter hyperbolic group $SO(1, 1)$ acting on the channel vector,

$$B(t) = \begin{pmatrix} e^{-\Gamma_\tau t} & 0 \\ 0 & e^{+\Gamma_\tau t} \end{pmatrix}, \quad \det B(t) = 1, \quad (14)$$

and $\alpha^{-1} = \xi\eta$ is its invariant. The coherent channel ξ decays at rate Γ_τ , the radiative channel η grows at rate Γ_τ , and their product — the Noether charge of the temporal fibre — is permanently locked. The individual channels are *not* conserved; they breathe in opposite directions. This is the difference between $\boldsymbol{\alpha}$ and the scalar one usually writes: α is the conserved norm of a living object, and the object itself has an orientation on Σ that the breathing dynamics continually rotates.

The irreversible advance of the logarithmic phase $\theta_{\log} = \frac{1}{2} \log(\eta/\xi)$ at rate $\dot{\theta}_{\log} = \Gamma_\tau$ is what a boundary observer calls time. The elementary phase quantum is $\pi\alpha$, the minimal resolvable phase step at the rank-5 boundary, below which two distinct bulk states project to the same observable because their difference lives in $\ker \Pi$. The reason the boundary supports exactly two channels and not three is information-theoretic: the entropic window of the rank-5 boundary is

$$S_\varphi = \frac{\varphi^4}{5} \approx 1.37 \text{ nat} \approx 1.98 \text{ bit}, \quad (15)$$

just enough for two independent binary distinctions and not three. The single Noether current produces the two-channel balance (7) — ξ for curvature, η for radiation — because the boundary has room for exactly two bits.

4 The rank and the static pole

The palindromic operator below carries the structural content of the projection but not yet its rank. The rank is fixed by a separate question — how many independent modes Π supports on $S^1(\tau)$ before the singular spectrum drops below threshold — and two independent derivations, one spectral and one algebraic, converge on the same integer.

4.1 Palindromic closure

Normalise $\hat{\xi} := \sqrt{\alpha} \xi$, $\hat{\eta} := \sqrt{\alpha} \eta$, so $\hat{\xi}\hat{\eta} = 1$. The ratio $x = \hat{\xi}/\hat{\eta}$ satisfies $\lambda^2 - k\lambda + 1 = 0$ with $k = x + x^{-1}$, whose unit constant term encodes $\hat{\xi}\hat{\eta} = 1$. This palindromic structure is the boundary image of Majorana self-conjugacy: charge conjugation maps $x \leftrightarrow x^{-1}$. The Pincherle dilatation $D = \tau\partial_\tau$ compresses under projection to $D_{\text{eff}} = \Pi D \Pi^*$, inheriting the symmetry,

$$D_{\text{eff}}^2 - k D_{\text{eff}} + 1 = 0, \quad (16)$$

and all subsequent structural results follow from (16) together with the rank.

4.2 Rank locking

Lemma 1 (Geometric and algebraic derivations converge on $r = 5$). *The rank is fixed twice, from opposite directions.*

- Geometric (spectral threshold). Equation (16) forces $\kappa = \sqrt{5}$. Mode n contributes singular value $\sigma_n = \kappa^{W-n}$, surviving the unit threshold iff $\sigma_n \geq 1$. With $\sigma_5 = \kappa^0 = 1$ and $\sigma_6 = \kappa^{-1} < 1$, exactly five modes survive: $W = 5$.
- Algebraic (Fibonacci fixed point). Self-conjugacy forces $|O_n|^2 = F_n$ (Proposition 1); the fixed-point condition $F_{n^*} = n^*$ has unique non-trivial solution $n^* = 5$.

Both give $r = 5$. Their agreement is the rank-locking: the unique integer at which the spectral geometry of the projection and the algebra of the Majorana field coincide.

The threshold check is a verification, not a definition. The admissible catalogue follows from $\pi(n) \leq r$ with π the prime-counting function: $N_{\text{adm}} = \{2, 3, 5, 7, 11\}$. The mode $n_{\text{conf}} = 7$ is confined — it does not factorise within the Fibonacci basis $\{2, 3, 5\}$ and lives in $\ker \Pi$.

4.3 The transfer function: Pincherle dilatation of the autosimilar level

The rank $r = 5$ does not yet give the pole; for that we need the bulk-to-boundary scale, and this is where the Pincherle dilatation does the work rather than being assumed. The generator of multiplicative rescaling on the fibre is

$$\mathcal{D} = x \frac{d}{dx} = \frac{d}{d(\log x)}, \quad \mathcal{D}(x^s) = s x^s, \quad (17)$$

whose eigenfunctions are the Mellin characters x^s . Setting $y = \log x$, $u = \log \varphi$, the autosimilar lattice $\{x_k = \varphi^k\}$ becomes the arithmetic lattice $\{ku\}$, and $\mathcal{D}$ acts as d/dy with the Jordan ladder $\mathcal{D}(y^n) = ny^{n-1}$. Since $\mathcal{D}$ lowers degree by one, the unique five-dimensional downward-closed $\mathcal{D}$ -invariant subspace containing constants is the log-cumulant space

$$V_5 = \text{span}\{(\log x)^n\}_{n=0}^4, \quad (18)$$

the polynomial jets of the Mellin character ($\partial_s^n x^s = (\log x)^n x^s$); modes $n \geq 5$ lie in $\ker \Pi$. This fixes the numerator without freedom: evaluating the projected kernel at the base point $x_1 = \varphi$ gives

$$P_4(z) = \sum_{n=0}^4 (\log \varphi)^n z^n, \quad z_k := \varphi^{-k}, \quad (19)$$

each coefficient the n -th power of the additive lattice step $u = \log \varphi$.

The pole is fixed by the bulk-to-boundary scale Λ , derived in three steps from (17). *Step 1:* the quadratic autosimilar level is the monomial x^2 ; the elementary bulk scale increment is the Pincherle dilatation of this level evaluated at $x = \varphi$,

$$\mathcal{D}(x^2)|_{x=\varphi} = 2\varphi^2. \quad (20)$$

Step 2: the spectral curvature of the Fibonacci recursion is $\kappa = \sqrt{5}$, which the identity $\sqrt{5} = 2(\varphi - \frac{1}{2})$ expresses in the autosimilar variable. *Step 3:* the bulk-to-boundary projection scale is the product of the Pincherle increment and the spectral curvature,

$$\Lambda := \mathcal{D}(x^2)|_{x=\varphi} \cdot \kappa = 2\varphi^2 \sqrt{5} = 4\varphi^2 \left(\varphi - \frac{1}{2}\right). \quad (21)$$

No coefficient is inserted: Λ is the Pincherle dilatation of the quadratic level times the spectral curvature. Using $\varphi^2 = \varphi + 1$ it expands in $\mathbb{Q}(\sqrt{5})$ as

$$\Lambda = 4\varphi^2 \left(\varphi - \frac{1}{2}\right) = 5 + 3\sqrt{5}, \quad (22)$$

and only here does the Fibonacci form appear, as a *consequence*: with $F_5 = 5$, $F_4 = 3$ one has $\Lambda = F_5 + F_4 \sqrt{5}$. The integers 5 and 3 are not chosen to land on a number — they are read off from $2\varphi^2 \sqrt{5}$ after the fact.

The numerator is not generic: the rank-five jet space V_5 is not closed under multiplication, so the scalarisation is the unique unital filtered-algebra homomorphism $\mathcal{S}_h : \mathbb{R}[y]/(y^5) \rightarrow \mathbb{R}[z]/(z^5)$ with $\mathcal{S}_h(y) = hz$, where $h = \log \varphi$ is the unique positive real with $e^h + e^{-h} = \sqrt{5}$ (the Clifford curvature of the class). This gives $\mathcal{S}_h(y^n) = h^n z^n$ and fixes $P_4(z) = \sum_{n=0}^4 h^n z^n$ with $h = \log \varphi$. The projection is bulk-to-boundary, and the observable coupling is the scale applied twice (the norm of the projected projection), so the single pole of the filter sits at

$$H_\alpha(z) = \frac{P_4(z)}{1 - \alpha z}, \quad \boxed{\alpha_{\text{QGT}}^{-1} = \Lambda^2 = (2\varphi^2\sqrt{5})^2 = (5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5} = 20\varphi^4 \approx 137.082.} \quad (23)$$

The closed form $20\varphi^4$ is exact: $(5 + 3\sqrt{5})^2 = 70 + 30\sqrt{5}$ and $20\varphi^4 = 20(\varphi + 1)^2 = 20(3\varphi + 2) = 70 + 30\sqrt{5}$ using $\varphi^2 = \varphi + 1$. The same constant carries a Hilbert–Cayley reading that makes the single pole manifest: the rational form

$$A_H(\lambda) = \frac{1}{20} \frac{(\lambda - 1)^2}{\lambda + 1} \quad (24)$$

has a double zero at $\lambda = 1$ and a simple pole at $\lambda = -1$, and evaluated at the golden point $A_H(\varphi) = \varphi^{-2}/(20\varphi^2) = 1/(20\varphi^4) = \alpha_{\text{QGT}}$. The double zero is the two retained chiral channels, the simple pole the single irreducible kernel orbit; this is the structural origin of the one-pole filter $H_\alpha(z)$, whose pole sits at $z = \alpha_{\text{QGT}}^{-1}$. The chain is a derivation, each arrow either a computation or a proved proposition,

$$S^1(\tau) \Rightarrow \partial_I J^I = 0 \Rightarrow \Xi = C\bar{\Xi}, (\gamma_6)^2 = 5 \Rightarrow r = 5 \Rightarrow \mathcal{D}, V_5 \Rightarrow \Lambda = 2\varphi^2\sqrt{5} \Rightarrow \alpha_{\text{QGT}}^{-1} = \Lambda^2 = 20\varphi^4, \quad (25)$$

with no number inserted at any step: α is assigned only at the last arrow, after Λ^2 is computed. The full Pincherle derivation of the filter, the boundary metric, and the rank-five SVD characterisation are in the companion study [9]. The root matrix is the regular representation of Λ in $\mathbb{Z}[\sqrt{5}]$,

$$M = \begin{pmatrix} F_5 & \kappa^2 F_4 \\ F_4 & F_5 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 3 & 5 \end{pmatrix}, \quad \det M = -20, \quad \|M^2\|_{\mathbb{Z}[\sqrt{5}]} = \alpha_{\text{QGT}}^{-1}. \quad (26)$$

This single matrix carries both observables of the construction. Its squared norm is α^{-1} ; its diagonal contraction will give $\sin^2 \theta_W$ in Section 6. The off-diagonal entry encodes the gauge-class count $|\mathcal{C}_{\text{obs}}| = 3$, and the closure

$$\frac{c_{\text{EW}} + n_{\text{conf}}}{\kappa^2} = \frac{13 + 7}{5} = 4 = \dim \mathbb{R}^4, \quad c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 13, \quad (27)$$

shows the electroweak and colour sectors together covering exactly the four observable dimensions. The coupling tensor (13) is the dynamical face of M : the same root matrix, read as a static norm, gives α^{-1} ; read as a living hyperbolic flow on Σ , gives α .

5 Two structural corrections and the readout spectrum

The static pole (23) is the value of α^{-1} at the symmetric point $\xi = \eta$, with no boundary dynamics. It is the bare quantity. The measured value is the cycle average accessible to a low-energy observer, and two parameter-free effects separate the two: the holonomy residue of the confined mode $n = 7$, exact in $\mathbb{Z}[\sqrt{5}][\pi]$, and the breathing back-action, exact in $\mathbb{Z}[\sqrt{5}]$. Neither is a QED loop — they are structural, fixed by the projection geometry before any diagram is computed — and neither replaces the renormalisation group: together they fix the boundary input $\alpha^{-1} \approx 137.036$ at the atomic lock-in scale, from which QED running then proceeds. The atomic lock-in is identified by the Rydberg constant $R_\infty = m_e c \alpha^2 / (2h)$, which fixes the *readout* of the scale at $n^* = 5$; the electron mass entering it is the on-shell value $m_e^{\text{on-shell}}$, used as a physical anchor of the QED readout and not derived from the lock-in. Obtaining m_e from the channel pair (ξ, η) belongs to a separate sector and is not claimed here.

5.1 Holonomy correction

Lemma 2 ($n = 7$ is the unique first-order contribution; $c_{\text{EW}} = 13$ forced). *Among the admissible modes, $n = 7$ is the unique element of $\ker \Pi$ contributing at first order. The modes $\{2, 3, 5\}$ lie in $\text{Im } \Pi$ and generate no residue; $n = 11$ is the domain-wall saturation mode, not in $\ker \Pi$, and produces no holonomy residue. The bleed-through multiplicity is $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 13$: without Majorana self-conjugacy a doubled bookkeeping would count 26 channel copies; $\Xi = C\bar{\Xi}$ identifies each channel with its conjugate, reducing $26 \rightarrow 13$. The same 13 is the denominator of $\sin^2 \theta_W = 3/13$.*

Mode $n = 7$ winds on $S^1(\tau)$ with holonomy angle $\pi/7$; Π cannot absorb this winding since $7 \in \ker \Pi$, so the amplitude $\kappa \cdot (\pi/7) = \sqrt{5} \pi/7$ bleeds through the electroweak channels, reduced by the closed unit winding of the base mode (which lies in $\text{Im } \Pi$ and is invisible at the boundary):

$$\varepsilon_7 = \frac{\sqrt{5} \pi}{7} - 1 \in \mathbb{Z}[\sqrt{5}][\pi]. \quad (28)$$

The subtracted unit is structural, not an adjustable offset; the same rule $\varepsilon_n = \sqrt{5} \pi/n - 1$ applies to every counterfactual mode of Table 2. This correction shifts α_{QGT}^{-1} to within 4.4×10^{-5} of the measured value.

5.2 Breathing back-action, and the single-channel rate

Here the tensor structure does real work. Because the coupling is the two-component object (13), a low-energy QED measurement samples the step-averaged norm of α , not a bare scalar. The lock-in is anisotropic: $\xi^*/\eta^* = n_{\text{sat}}/n^* = 11/5$, the ratio of the saturation mode to the lock-in mode. The two channels present to the observer at effective rates

$$\omega_\xi = \Gamma_\tau \cdot \frac{n^*}{n_{\text{sat}}} = \frac{\varphi}{\sqrt{5}} \cdot \frac{5}{11}, \quad \omega_\eta = \Gamma_\tau \cdot \frac{n_{\text{sat}}}{n^*} = \frac{\varphi}{\sqrt{5}} \cdot \frac{11}{5}, \quad \omega_\xi \omega_\eta = \Gamma_\tau^2 = \frac{\varphi^2}{5}. \quad (29)$$

The cycle average of α^{-1} over the breathing orbit has the multiplicative form, not the additive one, because the oscillating quantity is a denominator: $\alpha^{-1}(t) = \alpha_0^{-1}/(1 - f_{\text{breath}} \sin^2 \omega t)$ gives

$$\langle \alpha^{-1} \rangle_{\text{cycle}} = \frac{\alpha_0^{-1}}{1 - f_{\text{breath}}} + O(f_{\text{breath}}^2), \quad \alpha_0^{-1} = \alpha_{\text{QGT}}^{-1} - c_{\text{EW}} \varepsilon_7, \quad (30)$$

the standard geometric-mean result for an oscillating denominator. The absolute scale is fixed by the filter hierarchy (Appendix D): the back-action of one breathing cycle at the observer level $k_{\text{obs}} = n_{\text{min}} = 2$ is the invariant product (29), suppressed by the scale hierarchy $S^{k_{\text{obs}}(1+n_{\text{min}})} = 10^6$ and renormalised by the Majorana coupling $\varphi = \kappa \Gamma_\tau$,

$$f_{\text{breath}} = \frac{\omega_\xi \omega_\eta}{\varphi \cdot S^6} = \frac{\varphi^2/5}{\varphi \cdot 10^6} = \frac{\varphi}{5 \times 10^6} = (1 + \sqrt{5}) \times 10^{-7} \in \mathbb{Z}[\sqrt{5}] \cdot 10^{-7}. \quad (31)$$

The factor that matters physically for the readout spectrum is the splitting of f_{breath} into channel rates. The pair acts on the locked product carrying the Majorana factor 2, so

$f_{\text{breath}} = 2\varphi \times 10^{-7} \text{ (pair, on the invariant),} \quad f_1 = \frac{1}{2}f_{\text{breath}} = \varphi \times 10^{-7} \text{ (single channel).}$

(32)

The distinction is not bookkeeping. f_{breath} is the rate at which the *pair* of channels modulates the conserved invariant $\xi\eta$; f_1 is the rate of a *single* channel breathing on Σ . A symmetric (self-)readout couples to the invariant and feels f_{breath} averaged to zero — it sits on the fixed point. An asymmetric (channel-weighted) readout couples to one channel and is displaced by f_1 . This is the quantitative content of the prediction announced in Section 3.7, and Section 5.4 confronts it with data.

5.3 Self-consistent fixed point

$$\alpha_{\text{CODATA}}^{-1} = \frac{\alpha_{\text{QGT}}^{-1} - c_{\text{EW}} \varepsilon_7}{1 - f_{\text{breath}}}. \quad (33)$$

Table 1: Numerical chain from static pole to CODATA 2022. No CODATA value is inserted; the final residual -6.73×10^{-9} is 0.32σ of the CODATA 2022 uncertainty (its conjectured structural form is discussed in Remark 1).

Step	α^{-1}	Residual from CODATA 2022
Static pole $(5 + 3\sqrt{5})^2$	137.082 039	4.6×10^{-2}
$-c_{\text{EW}}\varepsilon_7$	137.035 955	4.4×10^{-5}
$\div(1 - f_{\text{breath}})$	137.035 999 1703	-6.73×10^{-9}
CODATA 2022 [4]	137.035 999 177(21)	(reference)

Remark 1 (Conjectured structural form of the residual; falsifiable). The zero-parameter value $\alpha_{\text{QGT,phys}}^{-1} = 137.035\,999\,1703$ deviates from CODATA 2022 by

$$\delta_{\text{res}} = \alpha_{\text{QGT,phys}}^{-1} - \alpha_{\text{CODATA 2022}}^{-1} = -6.73 \times 10^{-9} = -0.32 \sigma, \quad (34)$$

inside the experimental uncertainty. I record an observation about its *structure*, kept deliberately separate from the derivation above and not used as a correction. The residual admits the closed form

$$\delta_{\text{res}}^{\text{struct}} = -\sqrt{M_{12} M_{21}} \cdot S^{-3(1+n_{\text{min}})} = -3\sqrt{5} \times 10^{-9}, \quad (35)$$

matching (34) to 0.3% (-6.708×10^{-9} against -6.73×10^{-9}). Both factors are objects already fixed by the construction. The coefficient $3\sqrt{5} = \sqrt{45} = \sqrt{M_{12}M_{21}}$ is the geometric mean of the two off-diagonal entries of the root matrix M of (26), i.e. an intrinsic invariant of M lying in $\mathbb{Z}[\sqrt{5}]$; it is the same $\mathbb{Z}[\sqrt{5}]$ -combination ($\kappa^2|\mathcal{C}_{\text{obs}}| = 15$ and $|\mathcal{C}_{\text{obs}}| = 3$) that governs the gauge sector. The scale $10^{-9} = S^{-3(1+n_{\text{min}})}$ is the filter suppression $S^{k(1+n_{\text{min}})}$ of Appendix D evaluated at the level $k = 3$ — the molecular/hadronic boundary of the cascade (39) — one level deeper than the atomic lock-in $k = 2$ that fixes the 10^{-7} scale of f_{breath} .

The status of (35) is conjectural (C), and I am explicit about what is and is not established. *Established*: the numerical coincidence and the fact that both the coefficient $\sqrt{M_{12}M_{21}}$ and the exponent S^{-9} are fixed by structure already present, with no new quantity introduced. *Not established*: why the residual should sit at the level $k = 3$ rather than $k = 2$. The placement is motivated — the residual is naturally read as the imprint of the first boundary level below the lock-in, the level the atomic readout does not resolve — but it is not derived, and (35) is therefore not promoted to a correction: the derived value remains 137.035 999 1703, and δ_{res} remains the deviation from experiment.

The conjecture is falsifiable, and the test is sharp. At present the residual sits at 0.32σ and is statistically indistinguishable from zero; the structural and statistical readings cannot yet be separated. They separate once the CODATA uncertainty falls below $\sim 3 \times 10^{-9}$, expected by the CODATA 2030 adjustment given the projected improvement in $g-2$ and recoil determinations. If δ_{res} is a structural constant, the central value will remain anchored at $-3\sqrt{5} \times 10^{-9}$ from $\alpha_{\text{QGT,phys}}^{-1}$ as the error bar shrinks; if it is a statistical fluctuation, it will drift toward zero. The construction commits to the former: a pre-assignable, parameter-free target for the next two CODATA adjustments.

Remark 2 (The numbers are discrete, not tuned). The agreement is not obtained by adjusting a continuous parameter — there is none. The cascade depends on three discrete structural choices: the confinement mode $n \in N_{\text{adm}}$, the electroweak multiplicity $c \in \{5, 8, 13, 21\}$ (the

only values with a structural derivation in the rank-5 filter), and the coupling form of the back-action (multiplicative on the corrected pole, or additive as a bare rate). Table 2 lists the natural counterfactuals; only the structurally forced combination ($n=7$, $c=13$, mult.) reaches 7×10^{-9} , and every alternative is worse by more than three orders of magnitude.

Table 2: Discrete sensitivity analysis. Rows varying n use the uniform residue $\varepsilon_n = \sqrt{5} \pi / n - 1$, all else fixed; the bare-rate row reads the back-action as an additive shift.

n	c	type	α^{-1}	error
7	13	mult. (this work)	137.035 999 1703	-6.73×10^{-9}
3	13	mult.	≈ 119.64	1.7×10^1
5	13	mult.	≈ 131.82	5.2
11	13	mult.	≈ 141.78	4.7
7	5	mult.	≈ 137.064	2.8×10^{-2}
7	8	mult.	≈ 137.053	1.7×10^{-2}
7	21	mult.	≈ 137.008	2.8×10^{-2}
7	13	bare-rate shift	$\approx 137.035\,955$	4.4×10^{-5}

5.4 The readout spectrum of the coupling tensor

This is where the tensor structure becomes an experimental statement rather than an interpretation. On the invariant surface $\Sigma : \xi\eta = \alpha^{-1}$ the breathing flow is the $SO(1,1)$ group (14), and the scalar $\alpha^{-1} = \xi\eta$ is its exact invariant. The channels themselves are not conserved; they breathe at the single-channel rate f_1 (32).

A measurement does not access the channel pair. It accesses a scalar *contraction* of α with an apparatus-dependent channel weight, and there are exactly two classes. *Self-readouts* couple symmetrically and read the invariant $\xi\eta$ itself; they must return the fixed point. The electron $g-2$ is the canonical self-readout — QED measuring its own coupling. CODATA, a global least-squares adjustment dominated by α -sensitive determinations that contract the invariant channel, is treated here as belonging to this class and used as a consistency reference, not an independent determination. *Channel-weighted readouts* couple asymmetrically. A photon-recoil determination compares a mass readout h/M against a length readout R_∞ — two different contractions of the same projection — and is therefore displaced from the fixed point by the single-channel rate f_1 , in a direction set by channel weights the present derivation does not fix.

The two most precise recoil determinations disagree with each other at 5.5σ [5, 6]; the most precise $g-2$ value [7] and CODATA [4] sit between them. Against the parameter-free fixed point 137.035 999 1703:

Both self-readouts sit on the fixed point within their own uncertainties. The two channel-weighted readouts straddle it, and their splitting coincides with the single-channel rate (32) at 0.06σ — with the paired rate f_{breath} excluded at 5.6σ . No parameter was adjusted: f_1 is fixed by the cascade of Section 5, independently of and prior to this comparison.

I state the epistemic status carefully, because this is the kind of claim that invites overreading. (i) The Rb–Cs tension predates this work; the comparison is a zero-parameter *retrodiction*, not a prediction. (ii) Only the *magnitude* of the splitting is derived. The individual positions of the two recoil readings relative to the fixed point are asymmetric, and the channel weight of a given atomic species is not derived here; the ordering within the recoil class is open. (iii) I do not read the Cs/Rb discrepancy as a literal spacetime variation of α . It is a channel-readout splitting: two recoil experiments probe different contractions of the same projected α -operator. A conventional unresolved experimental systematic remains the competing null explanation,

Table 3: Readout spectrum of α against the parameter-free fixed point. The two channel-weighted (recoil) readouts straddle the fixed point; their mutual splitting matches the single-channel rate f_1 at 0.06σ , with the paired rate f_{breath} excluded at 5.6σ .

Determination	class	$\alpha^{-1} - \alpha_{\text{fp}}^{-1}$	σ_{exp}
$g-2$ (2023) [7]	self	-4.3×10^{-9}	-0.29
CODATA 2022 [4]	self (aggregate)	$+6.7 \times 10^{-9}$	+0.32
Rb recoil (2020) [6]	channel	$+3.6 \times 10^{-8}$	+3.3
Cs recoil (2018) [5]	channel	-1.2×10^{-7}	-4.6
Rb–Cs splitting (measured)		$1.60(3) \times 10^{-7}$	
single-channel rate $f_1 = \varphi \times 10^{-7}$		1.618×10^{-7}	(0.06 σ)

and the metrological community treats it as such; atomic-clock limits on a temporal drift of α are consistent with this reading, since the invariant $\xi\eta$ does not drift and only the contraction depends on the channel. (iv) The reading is falsifiable: if improved recoil determinations converge to a single value, the channel-weighted class collapses and this interpretation fails; if they remain split on the scale f_1 , it is confirmed. A readout in a structurally distinct channel class — the ^{229}Th nuclear-clock transition, whose α -sensitivity is strongly enhanced — would provide a pre-assignable test.

6 The gauge structure as a contraction of the same matrix

The pair $(r, |\mathcal{C}_{\text{obs}}|) = (5, 3)$ that fixes α^{-1} also organises the dimensionless gauge sector. I am explicit about scope: what follows is the gauge *skeleton* — the group structure $U(1) \times SU(2) \times SU(3)$, the tree-level mixing angle, and the algebraic origin of the colour triplet at the level of representations. It does not give masses, widths, or running couplings, which require the full perturbative machinery. These are consistency contractions of the projection that fixes α^{-1} , not an independent derivation of Standard Model dynamics.

6.1 The electroweak mixing angle

The root matrix M has diagonal $(r, r) = (5, 5)$ and off-diagonal $(\kappa^2 |\mathcal{C}_{\text{obs}}|, |\mathcal{C}_{\text{obs}}|) = (15, 3)$. The gauge tensor on the two-dimensional dynamical sector is built from the scalar contractions of M , with the diagonal $r = 5$ weighting the longitudinal channel and the off-diagonal $|\mathcal{C}_{\text{obs}}|/2 = 3/2$ the transverse,

$$G = \beta \begin{pmatrix} r & 0 \\ 0 & |\mathcal{C}_{\text{obs}}|/2 \end{pmatrix} = \beta \begin{pmatrix} 5 & 0 \\ 0 & 3/2 \end{pmatrix}, \quad \sin^2 \theta_W = \frac{G_{22}}{\text{tr } G} = \frac{3}{13} \approx 0.23077. \quad (36)$$

Against the Z -pole value 0.23122 ± 0.00003 [8], the tree-level ratio $3/13$ differs by about 0.19%. This is the uncorrected boundary ratio, before the ξ -channel holonomy and breathing corrections — the same corrections that close the α^{-1} sector — are applied; I do not read the gap as a failure. The pair $(r, |\mathcal{C}_{\text{obs}}|) = (5, 3)$ fixes both α^{-1} and $\sin^2 \theta_W$ as contractions of the same M : they are not independent results, and the same two corrections close $\sin^2 \theta_W$ with no new parameter. With the geometric suppression factor $\Upsilon_{\text{geom}} = \alpha^{-1}/S^2 = \varphi^4/r \in \mathbb{Z}[\sqrt{5}]$ ($S = n_{\text{min}} \cdot r = 10$),

$$\sin^2 \theta_W^{\text{closed}} = \frac{\sin^2 \theta_W^{(\text{II})} (1 + \Upsilon_{\text{geom}} \cdot \frac{|\mathcal{C}_{\text{obs}}|}{n_{\text{conf}}} \cdot \varepsilon_7)}{1 - \Upsilon_{\text{geom}} f_{\text{breath}}} \approx 0.23125, \quad (37)$$

within 0.75σ of the Z -pole value. The same denominator $c_{\text{EW}} = 2r + |\mathcal{C}_{\text{obs}}| = 13$ that appears in $\sin^2 \theta_W = 3/13$ governs the holonomy correction to α^{-1} : the electroweak structure and the fine-structure constant are two scalar contractions of the one Fibonacci matrix M .

6.2 Electroweak channels and colour

The two chiral channels carry the $U(1) \times SU(2)$ sector with anisotropy $\xi^*/\eta^* = 11/5$ at the lock-in; Majorana self-conjugacy protects custodial symmetry $M_{W+} = M_{W-}$ as the palindromic symmetry of the mode-weight tensor (Appendix E). Colour comes from the confined mode: $n_{\text{conf}} = 7$ is the unique element of N_{adm} that does not factorise within $\{2, 3, 5\}$, so it is forced into $\ker \Pi$, where the minimal triplet $K_7 \cong \mathbb{C}^3$ closes under its three pairwise $SU(2)$ subalgebras to $\mathfrak{su}(3)_c$ (Appendix F). Confinement is a projective necessity, not a dynamical consequence: $[5]_q = 0$ in $TL_4(\varphi)$ truncates the $n = 7$ sector exactly at the boundary.

7 One prediction, and what remains open

The time quantum $\delta\varphi = \pi\alpha$. The minimum observable phase step on the projected boundary is

$$\delta\varphi = \pi\alpha = \frac{\pi}{\alpha^{-1}} \approx 2.296 \times 10^{-2} \text{ rad.} \quad (38)$$

The statement is structural: $\delta\varphi$ is set by the coupling itself, so one half-cycle π of the fibre contains $\alpha^{-1} \approx 137.04$ elementary phase steps — not an integer, and no integer property of this count is used anywhere. The time quantum is the boundary analogue of $\hbar$: the granularity of observable temporal evolution in phase units, the same structural object as the inverse coupling, read in different units. An experiment resolving temporal phase below $\delta\varphi$ would probe the projection-induced discretisation of the boundary clock, a floor structurally distinct from thermal or shot noise.

The gravitational channel. On the boundary the scalar pole is the norm of the two-channel object (13). The longitudinal component $\alpha_{\parallel} \propto \xi$ is the curvature/gravitational readout, the transverse $\alpha_{\perp} \propto \eta$ the radiative/electromagnetic one, with $\alpha_{\parallel}/\alpha_{\perp} = 11/5$ at the lock-in. A QED experiment probes $\alpha_{\perp}$; the scalar α^{-1} is the $\mathbb{Z}[\sqrt{5}]$ -norm of the full α . The identification of $\alpha_{\parallel}$ with the gravitational sector is a structural readout of the same projection; the full dynamical derivation requires obtaining R^5 from the projection structure and is open. The same non-invertibility yields an unconditional irreversibility: each projection cycle loses a fraction $\delta U = 1 - 3\pi/(2\sqrt{5})^{3/2} \neq 0$ of bulk norm to the kernel, and the inequality is a theorem of pure mathematics (Lindemann–Weierstrass: $3\pi = (2\sqrt{5})^{3/2}$ would require $\pi \in \mathbb{Q}(\sqrt{5})$, which is false), so the arrow of time is a structural output of $\ker \Pi \neq \{0\}$, not a separate postulate.

The cosmological outlook. As an outlook beyond the α -sector, the same rank-5 projection makes structural predictions for the CMB power spectrum [12]: $\Omega_b h^2 = \sqrt{5}/100 \approx 0.02236$ (palindromic Silk residue), $\Omega_c h^2 = 1/(5\varphi) \approx 0.12361$ (from the Fibonacci fixed point at $n^* = 5$), $\tau_{\text{reion}} = 7/130$ (from the confined mode $n_{\text{conf}} = 7 \in \ker \Pi$), and $H_0 \simeq 67.9 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (from the analytic-return suppression of the breathing equation). These are outputs of the same projection, not fits; their derivations live in [12] and are quoted here only as the falsifiable outlook of the structure.

What is genuinely open. Beyond the closed α -sector and the one-step electroweak matching of Section 6, the open extensions are the full fermionic transporter beyond the charged-lepton sector [13], QCD dynamics, the full scheme-dependent electroweak running $\sin^2 \theta_W(Q^2)$, and the gravitational sector. I list these as open because they require work not yet done, not for lack of intention.

Stated plainly: Sommerfeld’s constant is what a Majorana bulk cannot deliver across its own projection interface. The bulk carries a coherent phase around $S^1(\tau)$; the projection discards

part of that coherence into a physical kernel; what the kernel holds back exerts a return tension on the boundary; and that tension, stabilised by the Fibonacci structure of the rank-5 filter, fixes the conserved norm of the coupling tensor at $\alpha^{-1} = 137.036$. The quantity is two-component because the projection is non-invertible, and what we measure is a contraction of it.

Data availability

A standalone verification script (`alpha_check.py`, 30-digit arithmetic) asserts every exact identity, the full cascade against CODATA 2022, each counterfactual of Table 2, the boundary/bulk duality gap, and the readout-spectrum comparisons of Section 5.4. No new experimental data were generated; all numerical comparisons use the published values cited.

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A Non-invertibility forces a non-trivial kernel

The setting of Theorem 1: let $\mathcal{A}$ be the C^* -algebra of observables of a field theory on $\mathbb{R}^4$. An admissibility projection $E : \mathcal{A} \rightarrow \mathcal{A}_{\text{adm}}$ satisfies (i) $\mathcal{A}_{\text{adm}} \subset \mathcal{A}$ a subalgebra; (ii) $E \circ E = E$; (iii) $\ker E \neq \{0\}$; (iv) $\ker E$ physically eliminated, not gauge-redundant. Gauge fixing fails (iv); coarse-graining fails (iv); decoherence fails (ii); renormalisation fails (iv); algebraic QFT imposes (i) externally. The only internal possibility is an automorphism, which is isometric, hence $\ker \phi = \{0\}$, failing (iii). No internal mechanism satisfies all four, and the corollary on the non-derivability of α follows.

B Uniqueness of $H_\alpha(z)$

$H_\alpha(z) = P_4(z)/(1 - \alpha z)$ is the unique rational transfer function of D_{eff} satisfying (i) rank $r = 5$; (ii) the palindromic equation (16); (iii) a single spectral pole at $z = \alpha^{-1}$; (iv) the Noether normalisation $K_\tau = \alpha^{-1}$. *Numerator*: the Pincherle dilatation $D = \tau \partial_\tau$ has eigenbasis $\{x^k\}$; the canonical Mellin-diagonal basis adapted to the rank-5 boundary is the log-cumulant basis $v_m(x) = (\log x)^m$, $m = 0, \dots, 4$, which diagonalises D_{eff} on the rank-5 image; any other basis introduces off-diagonal coupling. This fixes $P_4(z) = \sum_{m=0}^4 (\log \varphi)^m z^m$, with coefficients set by the golden eigenvalue. The value of α^{-1} depends only on the denominator pole, not on the normalisation of P_4 . *Denominator*: (16) forces H self-reciprocal, hence a palindromic denominator; degree 0 has no pole (excluded by (iii)), degree ≥ 2 gives two poles $z_1 z_2 = \pm 1$ and two Noether invariants (excluded by (iv)), leaving the unique degree-1 denominator $(1 - \alpha z)$. *Pole*: the pole value is not assumed but supplied by the Pincherle-derived bulk-to-boundary scale (21), $\Lambda = 2\varphi^2\sqrt{5} = 5 + 3\sqrt{5}$, the projection applied twice giving $z_{\text{pole}} = \Lambda^2 = 70 + 30\sqrt{5}$; the conjugate $\Lambda_- = 5 - 3\sqrt{5}$ gives a value inside the unit disk, absorbed into the numerator zeros, so the unique dominant pole is $\Lambda^2 = \alpha_{\text{QGT}}^{-1}$.

C The boundary code subspace

The return tension has the form $V'(|\Psi|) \propto |\Psi|^{2/3}$, the exponent forced by the unique dimensional relation between bulk volume ($\sim L^3$) and boundary surface ($\sim L^2$): $L^2 = (L^3)^\beta$ gives $\beta = 2/3$, the same exponent as the breathing map $B_{k+1} = 2B_k^{2/3}$. At the fixed point Ψ^* the field is scale-covariant, $\Psi^*(ax) = a^s \Psi^*(x)$, so the Fréchet derivative of $|\Psi|^{2/3}\Psi$ acts as a Mellin multiplier (a scalar on each Mellin mode), not a mode-coupling kernel; the linearised tangent operator L_* is therefore Mellin-diagonal. A Mellin-diagonal operator preserves the span of any set of Mellin modes; the rank-5 projection retains the first five, $v_m(x) \propto (\log x)^m$, $m = 0, \dots, 4$, and L_* maps this span to itself, giving exactly five eigenmodes — the code subspace.

D First-principles derivation of f_{breath}

Two inputs, both exact. *Filter structure*: the observer level is $k_{\text{obs}} = n_{\text{min}} = 2$, the renormalisation depth $1 + n_{\text{min}} = 3$, and the boundary filter scale decade $S = n_{\text{min}} \cdot r = 10$, so the

total suppression is $S^{k_{\text{obs}}(1+n_{\text{min}})} = 10^6$. The observer level is one rung of the boundary cascade, which assigns each level a physical role:

$$\begin{aligned} k = 0 & : \text{projection pole (bulk zero-mode),} & k = 3 & : \text{molecular/hadronic boundary,} \\ k = 1 & : \text{electroweak attractor,} & k = 4 & : \text{saturation mode,} \\ k = 2 & : \text{atomic lock-in (QED readout),} \end{aligned} \tag{39}$$

with physical determinations of α made at the lock-in $k = 2$, the unique level where $\xi^*\eta^* = \alpha^{-1}$ first stabilises (fixed by the Rydberg readout R_∞ , the boundary reading of the canonical pseudoinverse at that level). *Channel rates*: from $\xi^*/\eta^* = 11/5$, the rates (29) give $\omega_\xi\omega_\eta = \Gamma_\tau^2 = \varphi^2/5$. The back-action per cycle is the product, suppressed by 10^6 and renormalised by the Majorana coupling $\varphi = \kappa\Gamma_\tau$ (the product of the self-conjugacy norm $\kappa = \sqrt{5}$ and the native rate $\Gamma_\tau = \varphi/\sqrt{5}$), giving (31): $f_{\text{breath}} = (1+\sqrt{5}) \times 10^{-7}$. The algebraic direction is fixed by the exact $\mathbb{Z}[\sqrt{5}]$ -identity $2\varphi/(5+3\sqrt{5}) = (5-\sqrt{5})/10$; the scale 10^{-7} follows without free parameters. An independent check: the global loop gain at the critical node is $f_{\text{gain}} = (2\sqrt{5})^{3/2}/(3\pi) \approx 1.0035$, marginally above threshold, with excess $\delta F = 1 - f_{\text{gain}}^{-1} \approx 3.45 \times 10^{-3}$ equal to the per-cycle unitarity loss $\delta_F = 1 - 3\pi/(2\sqrt{5})^{3/2}$ of the arrow of time — the same transcendence ($\pi \notin \mathbb{Q}(\sqrt{5})$) that forbids $\varepsilon_7 = 0$ keeps the breathing from closing.

E Electroweak modal-weight tensor

Rank- r non-invertibility decomposes $\rho_{\text{pre}} = r\beta$ into three boundary residues with coefficients $\{r, |\mathcal{C}_{\text{obs}}|, n_{\text{min}}\}/2 = \{5, 3, 2\}/2$, giving the modal-weight tensor $T_{\text{EW}} = \frac{\beta}{2} \begin{pmatrix} 7 & 3 \\ 3 & 3 \end{pmatrix}$ with chiral contractions $\omega_- = 5\beta$, $\omega_+ = 2\beta$, $\omega_0 = \frac{3}{2}\beta$ (pre-dimensional, not eigenvalues). The off-diagonal $3 = |\mathcal{C}_{\text{obs}}|$ encodes Majorana self-conjugacy; the custodial statement $\rho_{\text{sym}} = 1$ is the palindromic symmetry of the rank-5 algebra, protecting $M_{W+} = M_{W-}$ after the dimensional cascade. Converting the pre-dimensional contractions into physical masses requires the non-linear boundary closure, which remains open.

F Colour algebra from the confined triplet

Proposition 3. *The Lie closure of the three pairwise $SU(2)$ subalgebras on $K_7 \cong \mathbb{C}^3$ is $\mathfrak{su}(3)_c$.*

Proof. With Gell-Mann matrices $\{\lambda_1, \dots, \lambda_8\}$, set $h_{13} = (\lambda_3 + \sqrt{3}\lambda_8)/2$, $h_{23} = (-\lambda_3 + \sqrt{3}\lambda_8)/2$. The subalgebras $s_{12} = \text{span}\{\lambda_1, \lambda_2, \lambda_3\}$, $s_{13} = \text{span}\{\lambda_4, \lambda_5, h_{13}\}$, $s_{23} = \text{span}\{\lambda_6, \lambda_7, h_{23}\}$ nominally give nine generators; the Cartan elements overlap, $h_{13} + h_{23} = \sqrt{3}\lambda_8$ and $h_{13} - h_{23} = \lambda_3$, reducing the span to $\{\lambda_1, \dots, \lambda_8\}$. The commutators $[\lambda_1, \lambda_2] = 2i\lambda_3$, $[\lambda_4, \lambda_5] = 2ih_{13}$, $[\lambda_6, \lambda_7] = 2ih_{23}$, $[\lambda_1, \lambda_4] = i\lambda_7$ all close within this span, so $\dim = 8 = \dim \mathfrak{su}(3)_c$. This is the minimal colour skeleton; full QCD dynamics are not derived. $\square$